

A Report

on

Agentic-RAG Survey

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CERTIFICATE

This is to certify that the project entitled “**Agentic-RAG Survey**” is a bonafide work carried out as PBL-2 **End Term Assessment** in partial fulfillment for the award of the degree of Bachelor of Technology in Computer Science and Engineering, by **Harshit Mandad** bearing registration number **2427030400**, during the academic semester IV of year 2025-2026.

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ABSTRACT

Retrieval-Augmented Generation (RAG) has emerged as an effective approach to improve the factual reliability of Large Language Models (LLMs) by integrating external knowledge sources into the generation process. By combining retrieval systems with generative models, RAG enables responses to be grounded in verifiable information, thereby reducing hallucination and improving contextual relevance. However, traditional RAG architectures rely on static, single-pass retrieval mechanisms, which limit their effectiveness in handling complex, multi-step reasoning tasks and dynamically evolving information requirements.

As real-world applications increasingly demand systems capable of procedural reasoning, contextual adaptation, and goal-oriented execution, these limitations become more significant. Conventional RAG systems typically retrieve a fixed set of documents based on an initial query and generate responses without iterative refinement or structured task decomposition.

To address these challenges, Agentic Retrieval-Augmented Generation (Agentic RAG) has been introduced as an advanced paradigm that extends traditional RAG by incorporating autonomous planning, iterative retrieval, contextual memory, and tool-based reasoning. These systems enable the decomposition of complex queries into structured sub-tasks, allowing for dynamic refinement of retrieval strategies and improved reasoning accuracy. By leveraging intermediate feedback and contextual awareness, Agentic RAG systems can significantly reduce hallucination and enhance robustness in complex environments.

This work presents a comparative analysis of various Agentic RAG methodologies, including traditional retrieval-based systems, graph-based frameworks, multi-agent architectures, goal-oriented approaches, hierarchical models, adaptive systems, and domain-specific implementations. The analysis evaluates these methodologies across multiple dimensions such as reasoning capability, hallucination mitigation, computational cost, scalability, and system complexity. While advanced agentic approaches demonstrate improved performance in multi-step reasoning and contextual adaptation, they also introduce challenges such as increased computational overhead, architectural complexity, and the lack of standardized evaluation frameworks.

To overcome these limitations, this paper proposes a Reinforcement Learning (RL)-enhanced Agentic RAG framework that introduces adaptive, learning-driven optimization into the retrieval and reasoning pipeline. The proposed framework utilizes a hierarchical policy structure, where a high-level policy performs goal-oriented task decomposition and a low-level policy manages adaptive retrieval actions such as query reformulation and document selection.

Overall, this study highlights the importance of integrating adaptive learning mechanisms into Agentic RAG systems and demonstrates that reinforcement learning can serve as a promising direction for developing more reliable, scalable, and intelligent knowledge-grounded AI systems.

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1. Introduction

Large Language Models (LLMs) have significantly improved natural language understanding and generation, enabling applications such as question answering, summarization, and conversational AI. However, these models often suffer from issues such as hallucination, factual inaccuracies, and limited access to up-to-date or domain-specific information.

Retrieval-Augmented Generation (RAG) addresses these limitations by integrating external knowledge retrieval into the generation process, allowing responses to be grounded in reliable information. While this approach improves accuracy, traditional RAG systems rely on static, single-pass retrieval, making them less effective for complex queries that require multi-step reasoning.

To overcome these challenges, Agentic Retrieval-Augmented Generation (Agentic RAG) introduces capabilities such as iterative retrieval, task decomposition, and contextual reasoning. These systems enhance adaptability and improve response quality in complex scenarios. This work focuses on analyzing different Agentic RAG methodologies.

1.1 Motivation

The increasing use of AI systems in critical domains such as healthcare, finance, and scientific research has created a strong demand for reliable, context-aware, and accurate information generation. While Retrieval-Augmented Generation (RAG) has improved the factual grounding of responses, its traditional implementations are not well-suited for handling complex, multi-step reasoning tasks or dynamically evolving user queries.

In real-world scenarios, queries are often ambiguous, require reasoning across multiple sources, or involve sequential decision-making. Static retrieval approaches fail to adapt to such requirements, resulting in incomplete or inaccurate responses. Furthermore, existing Agentic RAG systems, although more advanced, largely depend on predefined rules and heuristics for decision-making, which limits their ability to learn from experience and optimize performance over time.

Another key motivation for this work is the lack of standardized evaluation frameworks and unified benchmarking methods for Agentic RAG systems. This makes it difficult to compare different methodologies and understand their effectiveness across various dimensions such as accuracy, scalability, and computational efficiency.

To address these challenges, this project is motivated by the need to develop a more adaptive and intelligent retrieval framework that can learn optimal strategies through interaction and feedback. By integrating reinforcement learning into the Agentic RAG pipeline, the system can dynamically refine its retrieval and reasoning processes, leading to improved performance, reduced hallucination, and enhanced scalability.

2. Literature Review

Retrieval-Augmented Generation (RAG) has emerged as a widely adopted approach to enhance the factual accuracy of Large Language Models (LLMs) by incorporating external knowledge retrieval into the generation process. Early research primarily focused on improving retrieval quality through dense vector search and hybrid retrieval techniques, enabling models to access relevant contextual information beyond their parametric knowledge.

Recent advancements have introduced the concept of Agentic Retrieval-Augmented Generation (Agentic RAG), which extends traditional RAG by incorporating autonomous reasoning, task decomposition, and iterative retrieval mechanisms. These systems enable models to break down complex queries into smaller sub-tasks, perform multi-step reasoning, and refine responses based on intermediate feedback.

Several methodologies have been proposed in this domain. Graph-based RAG systems utilize structured knowledge graphs to enable relational reasoning and improve interpretability. Multi-agent RAG architectures distribute tasks across multiple specialized agents, allowing parallel processing and improved performance in multi-hop reasoning tasks. Goal-oriented frameworks focus on structured planning and procedural reasoning, while hierarchical models introduce layered control mechanisms to manage complex workflows.

Adaptive Agentic RAG systems further enhance performance by incorporating feedback-driven learning mechanisms, allowing systems to refine retrieval strategies over time. Additionally, domain-specific RAG architectures leverage curated datasets and specialized knowledge bases to improve accuracy in fields such as healthcare, finance, and legal systems.

Despite these advancements, existing approaches face several challenges, including increased computational complexity, scalability issues, reliance on heuristic orchestration, and lack of standardized evaluation frameworks. These limitations highlight the need for more adaptive and learning-driven solutions.

2.1 Literature Review

TABLE 1 - Agentic-RAG Methodologies and Analysis

Approach	Key Idea	Advantages	Limitations
Traditional RAG	Single-pass retrieval + generation	Fast, simple, cost-efficient	No multi-step reasoning, limited adaptability
Graph-Based RAG	Uses knowledge graphs for reasoning	Structured reasoning, explainability	High complexity, scalability issues
Multi-Agent RAG	Multiple agents handle sub-tasks	Parallel execution, better reasoning	High cost, coordination issues
Goal-Oriented RAG	Task decomposition based on objectives	Strong multi-step reasoning	High latency, complexity
Hierarchical RAG	Multi-layer planning and control	Better workflow management	Error propagation, high cost
Adaptive RAG	Learns from feedback	Personalization, adaptability	Data dependency, instability
Domain-Specific RAG	Uses specialized datasets	High accuracy in specific domains	Poor generalization

2.2 Outcome of Literature Review

The literature review highlights that while Agentic RAG significantly improves reasoning capabilities and factual grounding compared to traditional RAG, several critical challenges remain. Most existing systems rely on predefined rules and heuristic-based decision-making, which limits their adaptability in dynamic environments.

Additionally, advanced architectures such as multi-agent and hierarchical systems introduce increased computational overhead and system complexity, making them difficult to scale. The lack of standardized evaluation frameworks further complicates the comparison of different methodologies.

A key insight from the review is that current systems lack learning-driven optimization, particularly in retrieval strategy selection and task decomposition. This creates a need for adaptive frameworks that can improve performance over time through interaction and feedback.

2.3. Problem Statement

Despite significant advancements in Retrieval-Augmented Generation and its agentic extensions, existing systems are limited by static retrieval strategies, heuristic-based orchestration, and lack of adaptive optimization mechanisms. These limitations result in reduced efficiency, increased hallucination, and poor performance in complex, multi-step reasoning tasks.

Furthermore, current Agentic RAG frameworks face challenges related to scalability, computational cost, and coordination complexity. The absence of learning-driven decision-making restricts their ability to adapt to dynamic queries and evolving knowledge environments.

Therefore, there is a need to develop an adaptive and scalable framework that can optimize retrieval strategies and task decomposition dynamically while maintaining accuracy and efficiency.

2.4 Research Objectives

The primary objectives of this project are:

- To analyze and compare different Agentic RAG methodologies based on performance, scalability, and reasoning capability
- To identify key limitations and research gaps in existing retrieval-augmented systems
- To design a framework to overcome a key limitation.
- To improve multi-step reasoning and reduce hallucination through feedback-driven optimization
- To enhance scalability and efficiency of retrieval systems in complex and dynamic environments

3. Methodology and Framework

This project follows a research-oriented methodology focused on analyzing existing Agentic Retrieval-Augmented Generation (Agentic RAG) systems and proposing an enhanced framework using Reinforcement Learning (RL). The methodology is divided into two main phases: comparative analysis and framework design.

In the first phase, different RAG methodologies such as traditional, graph-based, multi-agent, goal-oriented, hierarchical, adaptive, and domain-specific approaches are studied and compared. This analysis is performed based on key factors including reasoning capability, hallucination control, computational efficiency, and scalability.

Based on the insights obtained from the literature review, the second phase focuses on designing an RL-enhanced Agentic RAG framework. The proposed framework introduces adaptive decision-making into the retrieval and reasoning pipeline by integrating reinforcement learning with agentic components.

The system operates through a structured workflow in which a user query is first analyzed and decomposed into smaller sub-tasks. These sub-tasks are processed iteratively using retrieval and reasoning mechanisms. Reinforcement learning is then used to optimize retrieval strategies based on feedback such as accuracy, relevance, and efficiency.

This approach enables dynamic refinement of retrieval actions and improves the overall performance of the system in handling complex and multi-step queries.

3.1 System Architecture

The proposed system architecture consists of three primary components: the retriever module, the generator module, and the augmentation module, along with a reinforcement learning layer that enables adaptive optimization.

- **Retriever Module**

The retriever is responsible for fetching relevant information from external knowledge sources. Unlike traditional systems, it performs iterative and context-aware retrieval, refining queries based on intermediate results.

- **Generator Module**

The generator, typically a Large Language Model (LLM), synthesizes responses using both the input query and retrieved information. It also assists in generating intermediate reasoning steps and sub-tasks.

- **Augmentation Module (Knowledge Base)**

This module contains external data sources such as document repositories, databases, or knowledge graphs. It provides factual grounding and domain-specific information.

- **Reinforcement Learning Layer**

The RL layer introduces adaptive decision-making into the system. It optimizes retrieval strategies and task decomposition through feedback-driven learning.

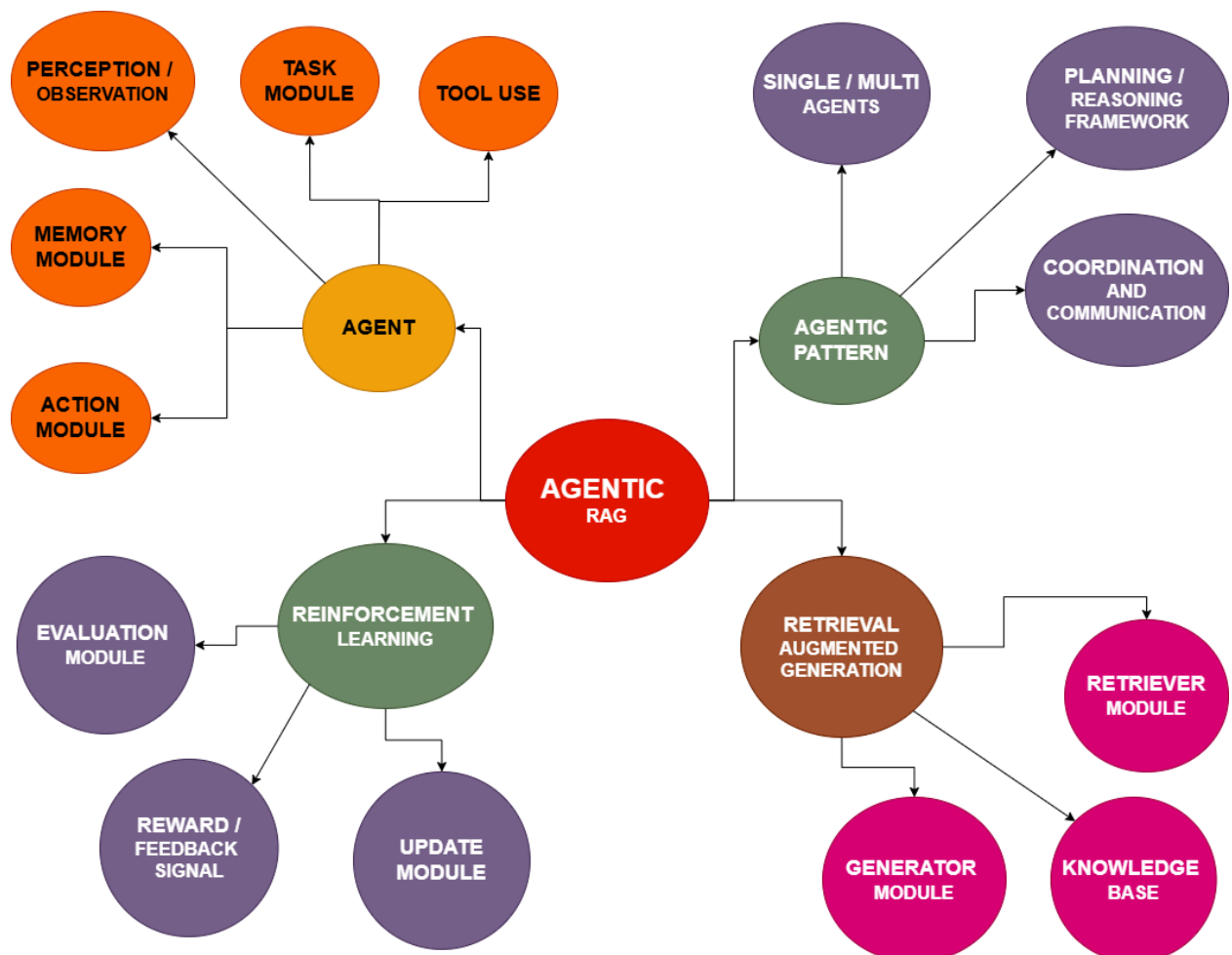


FIG. 1 - Agentic-RAG Framework

The interaction between these components enables the system to perform multi-step reasoning, iterative retrieval, and dynamic adaptation. The architecture ensures that the system continuously improves its performance based on feedback and environmental conditions.

3.2 Algorithms and Techniques

The proposed framework incorporates several key algorithms and techniques:

- **Reinforcement Learning (RL):** Reinforcement Learning is used to enable adaptive decision-making in the system. The framework follows a policy-based learning approach where the system learns optimal actions through interaction with the environment.
 - ❖ State: Represents the current query, context, and retrieved information
 - ❖ Action: Includes query reformulation, document selection, and retrieval decisions
 - ❖ Reward: Based on accuracy, relevance, reasoning quality, and efficiency
- **Hierarchical Reinforcement Learning:** The system utilizes a hierarchical structure consisting of:
 - ❖ High-Level Policy: Responsible for task decomposition and planning
 - ❖ Low-Level Policy: Handles retrieval actions and execution
- **Iterative Retrieval:** Unlike static retrieval, the system performs multiple retrieval steps, refining queries based on intermediate results. This improves accuracy and relevance.
- **Context-Aware Retrieval:** The system uses contextual information from previous steps to guide retrieval, ensuring that results are relevant to the overall objective.
- **Multi-Step Reasoning:** The framework supports reasoning across multiple stages, enabling it to solve complex queries that require combining information from different sources.

3.3 Detailed Design Methodologies

The detailed design of the proposed system follows a step-by-step workflow:

1. **Query Input:** The user provides an initial query to the system.
2. **Task Decomposition:** The high-level policy divides the query into smaller sub-tasks.
3. **Iterative Retrieval:** The low-level policy retrieves relevant documents for each sub-task and refines queries as needed.
4. **Context Integration:** Retrieved information is combined with previous context to maintain continuity.
5. **Response Generation:** The generator produces intermediate and final outputs based on retrieved data.
6. **Feedback and Reward Evaluation:** The system evaluates performance using metrics such as accuracy, relevance, and efficiency.
7. **Policy Update:** Reinforcement learning updates the policies to improve future decisions.

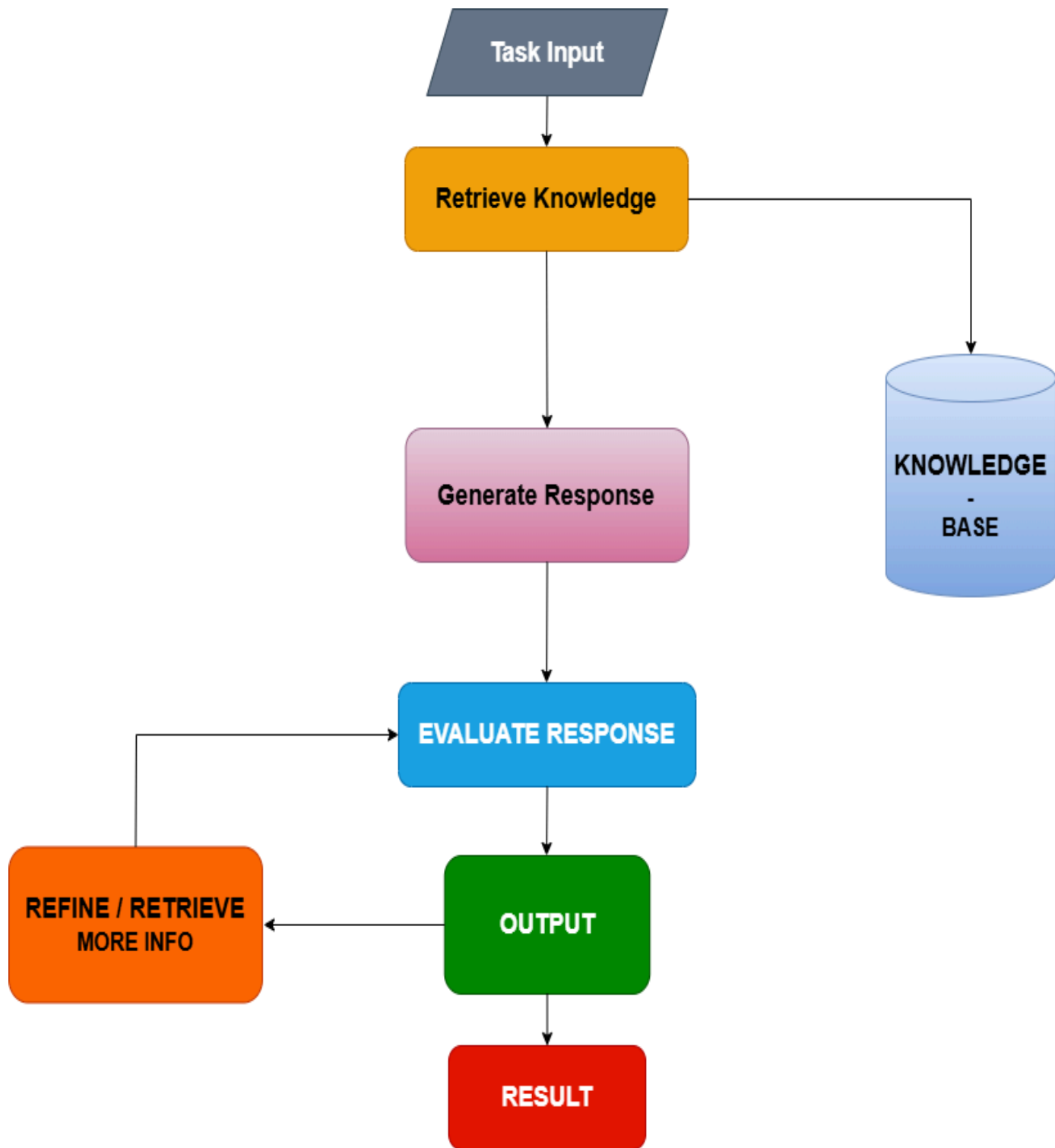


Fig. 2 - RAG Structural Flow

4. Work Done

The work carried out in this project primarily focuses on the analysis, design, and conceptual development of an enhanced Agentic Retrieval-Augmented Generation (Agentic RAG) framework integrated with Reinforcement Learning (RL).

Initially, an extensive study of existing Retrieval-Augmented Generation techniques and their agentic extensions was conducted. Various methodologies, including traditional RAG, graph-based RAG, multi-agent systems, goal-oriented frameworks, hierarchical architectures, adaptive models, and domain-specific implementations, were systematically analyzed.

Based on this analysis, a structured comparative framework was developed to evaluate these methodologies across key parameters such as reasoning capability, hallucination mitigation, computational efficiency, scalability, and system complexity. The findings were organized into tabular formats to clearly highlight strengths, limitations, and trade-offs.

Following the comparative study, a novel RL-enhanced Agentic RAG framework was proposed. The framework introduces a hierarchical reinforcement learning structure consisting of a high-level policy for task decomposition and a low-level policy for adaptive retrieval control. The design incorporates iterative retrieval, context-aware reasoning, and feedback-driven optimization mechanisms.

Additionally, system architecture diagrams, workflow models, and conceptual design structures were developed to illustrate the functioning of the proposed framework. The entire work was documented in the form of a research survey paper, aligning with academic and technical standards.

4.1 Results and Discussion

Although the proposed framework has not been implemented as a fully functional system, the analysis and design demonstrate significant potential improvements over existing approaches.

The comparative study reveals that traditional RAG systems are efficient but lack adaptability and multi-step reasoning capabilities. Advanced agentic frameworks improve reasoning performance but introduce challenges such as increased computational cost, complexity, and scalability issues.

The proposed RL-enhanced Agentic RAG framework addresses these limitations by introducing adaptive learning into the retrieval and reasoning process. The use of hierarchical reinforcement learning enables dynamic task decomposition and optimized retrieval strategies, which can improve accuracy and reduce hallucination.

Furthermore, the framework supports multi-step reasoning and iterative refinement, making it more suitable for handling complex and ambiguous queries. The integration of

feedback-driven learning allows the system to continuously improve its performance over time.

However, the approach also introduces challenges such as increased computational overhead, training complexity, and the need for well-designed reward mechanisms. These factors must be carefully considered during implementation.

Overall, the results of the study indicate that reinforcement learning can play a significant role in enhancing the capabilities of Agentic RAG systems and represents a promising direction for future research.

4.2 Individual Contribution

This project was carried out individually. The entire work, including literature review, comparative analysis, framework design, system architecture development, and report preparation, was completed by the author.

The author under the guidance of his mentor was responsible for:

- Conducting an in-depth study of Retrieval-Augmented Generation and Agentic RAG methodologies
- Designing the comparative analysis framework and tabular evaluation
- Proposing the Reinforcement Learning-enhanced Agentic RAG architecture
- Developing system design diagrams and workflow representations
- Writing and structuring the research survey paper and project report

TABLE 2 - Common Limitations

Limitations Identified
Scalability Optimization : Agentic architectures significantly increase computational complexity, limiting deployment in large-scale systems.
Learning-Driven Retrieval Policy : While adaptive and RL-based approaches show promise, stable and efficient policy learning remains underexplored.
Lack of Standardized Evaluation Framework : There is no unified benchmarking methodology for measuring agentic autonomy, goal completion, and hallucination control across architectures.

5. Conclusion and Future Plan

This project presented a comprehensive analysis of Retrieval-Augmented Generation (RAG) and its evolution into Agentic Retrieval-Augmented Generation (Agentic RAG). Through a structured comparative study of multiple architectures—including traditional, graph-based, multi-agent, goal-oriented, hierarchical, adaptive, and domain-specific approaches—the work identified key strengths as well as critical limitations in current systems.

The analysis shows that while Agentic RAG significantly enhances reasoning capability, contextual understanding, and multi-step task execution, most existing approaches rely on heuristic or rule-based strategies. This limits their adaptability, scalability, and efficiency in handling dynamic and complex real-world scenarios. Issues such as computational overhead, system complexity, and lack of standardized evaluation frameworks further highlight the need for more advanced solutions.

To address these challenges, this project proposed a Reinforcement Learning (RL)-enhanced Agentic RAG framework. The proposed system introduces a hierarchical learning mechanism consisting of a high-level policy for task decomposition and a low-level policy for adaptive retrieval control. By incorporating feedback-driven optimization, the framework enables dynamic refinement of retrieval strategies, improved reasoning accuracy, and reduced hallucination. Although the work is conceptual in nature, it establishes a strong foundation for developing intelligent, adaptive, and scalable retrieval systems.

Looking ahead, the future scope of this work lies in practical implementation and validation of the proposed framework. A key next step would be to develop a prototype system by integrating reinforcement learning algorithms with existing large language models and retrieval pipelines. This would allow evaluation on real-world datasets and provide insights into system performance, efficiency, and scalability.

Further research can focus on designing advanced reward functions that balance multiple objectives such as accuracy, relevance, latency, and computational cost. Improving scalability through distributed architectures, efficient retrieval mechanisms, and lightweight models will also be essential for real-world deployment. Additionally, incorporating human-in-the-loop feedback can enhance system reliability and alignment with user expectations.

The proposed framework can also be extended to domain-specific applications such as healthcare, finance, and education, where accurate and context-aware information retrieval is critical. Finally, the development of standardized evaluation metrics and benchmarking frameworks will be important for comparing different Agentic RAG systems and measuring their effectiveness.

Overall, this work highlights the potential of combining reinforcement learning with Agentic RAG to build more adaptive, efficient, and intelligent systems capable of robust reasoning in complex and evolving environments.

6. Outcome

- A comprehensive research survey paper on Agentic Retrieval-Augmented Generation (Agentic RAG) was developed, including comparative analysis and a proposed RL-enhanced framework.
- The work highlights key limitations in existing systems and presents a structured, adaptive solution for improving retrieval and reasoning.
- The study contributes to ongoing research in intelligent, scalable, and autonomous AI systems, with potential for future implementation and publication.

7. References

- [1] Zhang, Y., et al. (2025). “Agentic retrieval-augmented generation: A survey on agentic RAG.” arXiv.
- [2] Kumar, S., et al. (2025). “Retrieval-augmented generation (RAG) for fintech: Agentic design and evaluation.” arXiv.
- [3] Kumar, A., et al. (2025). “Agentic AI: Autonomous intelligence for complex goals—A comprehensive survey.” IEEE Access.
- [4] Li, X., et al. (2025). “From generative AI to agentic AI: Foundations, capabilities, and future directions.” arXiv.
- [5] (2025). “Domain-specific LLMs, retrieval-augmented generation and agentic AI: A unified architecture for specialized intelligence.” SSRN.
- [6] Singh, A. (2025). “Adaptive agentic RAG systems for improving adaptability and performance in AI-driven information retrieval.” SSRN.
- [7] (2025). “Efficient and transferable agentic knowledge graph RAG via reinforcement learning.” arXiv.
- [8] Li, Y. (2018). “Deep reinforcement learning: An overview.” arXiv.
- [9] Kaelbling, L. P., Littman, M. L., and Moore, A. W. (1996). “Reinforcement learning: A survey.” Journal of Artificial Intelligence Research.
- [10] Zhang, K., Yang, Z., and Basar, T. (2021). “A survey on multi-agent deep reinforcement learning.” Artificial Intelligence Review.
- [11] (2025). “A comprehensive survey on reinforcement learning-based agentic search: Foundations, roles, optimizations, evaluations, and applications.” arXiv.
- [12] Luo, H., et al. (2025). “Graph-R1: Towards agentic GraphRAG framework via end-to-end reinforcement learning.” arXiv.
- [13] Zheng, Q., et al. (2025). “End-to-end agentic RAG system training for traceable diagnostic reasoning.” arXiv.
- [14] Yoo, H. K. (2025). “Agentic RAG: Reinforcement learning-based insight-centric search generation.” KoreaScience.
- [15] Bagan, N. (2025). “Optimizing agentic retrieval-augmented generation using reinforcement learning-based methods.” Leiden Institute of Advanced Computer Science (LIACS).
- [16] Nagori, A., et al. (2025). “Open-source agentic hybrid RAG framework for scientific literature review.” arXiv.

8. Appendix : Survey Paper

Agentic Retrieval-Augmented Generation: A Comparative Survey

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Abstract—Retrieval-Augmented Generation (RAG) has emerged as an effective approach for improving the factual reliability of large language models (LLMs) by integrating external knowledge sources into the generation process. However, most conventional RAG systems rely on static, single-step retrieval, which limits their ability to handle complex queries and evolving information contexts. Agentic Retrieval-Augmented Generation (Agentic RAG) addresses these limitations by introducing capabilities such as iterative retrieval, task decomposition, contextual memory, and autonomous decision-making. These systems are able to dynamically refine queries, adapt retrieval strategies, and structure multi-step reasoning processes more effectively. In this work, we present a comparative study of prominent Agentic RAG paradigms, including goal-oriented, multi-agent, graph-based, and adaptive approaches. The analysis highlights their respective strengths in improving reasoning accuracy and reducing hallucinations, while also identifying practical challenges such as increased computational cost, system complexity, and limited scalability. To address these issues, we propose a reinforcement learning-based Agentic RAG framework that enables adaptive retrieval strategy selection through hierarchical policy learning. The proposed approach aims to improve decision-making, enhance adaptability, and support more reliable knowledge-grounded generation in complex environments. This paper provides a structured comparative analysis and highlights key research challenges in Agentic RAG systems.

Index Terms : Agentic RAG, Retrieval-Augmented Generation, Large Language Models (LLMs), Reinforcement Learning, Multi-Agent Systems, Adaptive Retrieval, Hallucination Mitigation

I. Introduction

Recent advances in Large Language Models (LLMs) have enabled significant progress in natural language understanding and generation. However, despite their capabilities, these models often generate factually incorrect or unsupported information, particularly when operating beyond their training data. This limitation poses serious challenges in domains where accuracy and reliability are critical. Retrieval-Augmented Generation (RAG) has been proposed as a practical solution to these challenges by combining information retrieval with text generation. By grounding responses in external knowledge sources, RAG improves factual consistency and reduces unsupported outputs. However, traditional RAG systems are inherently limited by their static design. Most approaches rely on a single retrieval step followed by generation, without any

mechanism for iterative refinement or reasoning. This makes them less effective for complex queries that require multi-step inference or contextual adaptation. As application demands grow—particularly in domains such as healthcare, finance, and education—there is an increasing need for systems that can not only retrieve relevant information but also plan, adapt, and reason dynamically. Overall, this work aims to bridge the gap between static retrieval architectures and adaptive, learning-driven systems by introducing reinforcement learning into the Agentic RAG paradigm.



Fig. 1: Basic Components of RAG

Agentic Retrieval-Augmented Generation (Agentic RAG) has recently emerged as a promising extension to address these challenges. By incorporating planning, memory, and tool usage, these systems enable structured task execution and iterative reasoning. Despite these improvements, many existing Agentic RAG frameworks still rely on heuristic or rule-based coordination strategies, which limits their adaptability in dynamic environments. Motivated by these limitations, this work explores the integration of reinforcement learning into Agentic RAG to enable adaptive and data-driven decision-making. In addition, a comparative analysis of existing methodologies is conducted to better understand their strengths, limitations, and trade-offs.

II. Overview of Agentic Retrieval-Augmented Generation

Agentic Retrieval-Augmented Generation (Agentic RAG) extends traditional RAG architectures by introducing elements of planning, adaptability, and autonomous decision-making. Unlike conventional pipelines that follow a fixed retrieve-then-generate structure, Agentic RAG systems incorporate iterative

Methodology	Limitations/Challenges
Multi-Agent Agentic RAG	- Multi-Agent Coordination and Communication - Response Conflict across agents
Goal-Oriented Agentic RAG	- Scalability issue - No standard evaluation metric system - Risk of API cost/tool misuse
Traditional Agentic RAG	- Poor performance for complex tasks - No adaptability/learning - Limited knowledge base - Dependency on graph quality
Graph-Based Agentic RAG	- Integration Complexity - Struggle with newly-emerging info - More layers=Scalability issues
Hierarchical RAG	- Lack of standard evaluation metrics - High computational cost - Lack of standardized benchmark
Capability-Centric Agentic RAG (Conceptual)	- High integration complexity - Data dependency for learning
Adaptive Agentic RAG	- Stability and drift issues - Evaluation Complexity - Over-Correction risk - Evaluation Benchmark gaps
Corrective Agentic RAG	- No universal way to measure answer correctness
Domain-Specific Agentic RAG	- Data availability constraint - Knowledge drift
Unified-domain Agent Stack	- Scalability issues - Lack of standardized evaluation framework

TABLE II: Identified Limitations Across Agentic-RAG Methodologies

prone to error propagation across layers. The added architectural complexity and computational overhead further limit scalability and real-time applicability.

G. Adaptive Agentic RAG

Adaptive Agentic RAG systems incorporate feedback-driven learning mechanisms to dynamically refine retrieval strategies and improve performance over time. These systems demonstrate strong adaptability and personalization capabilities. Despite these benefits, adaptive approaches introduce higher computational overhead, longer response times, and dependency on high-quality feedback data. Additionally, they face challenges in evaluation due to dynamic system behavior and risk overfitting to repetitive query patterns, potentially reducing generalization.

H. Domain-Specific and Unified Agentic Architectures

Domain-specific Agentic RAG systems leverage curated knowledge bases to achieve high accuracy and reduced hallucination in specialized domains such as healthcare, legal, and finance. Unified-domain architectures extend this concept by

integrating domain-adapted LLMs with retrieval and agentic components for end-to-end reliability. However, these systems face limitations in generalization beyond their target domain and require significant effort in data curation and maintenance. Knowledge drift, high system complexity, and lack of standardized benchmarking frameworks further complicate their scalability and evaluation.

I. Identified Research Gaps

The identified limitations and research gaps across different Agentic RAG paradigms are summarized in Table III, providing a consolidated view of current challenges. While existing approaches improve reasoning and retrieval capabilities, they often do so at the cost of increased complexity and computational overhead. In addition, there is a lack of standardized evaluation frameworks, making it difficult to compare different systems objectively. These challenges highlight the need for more adaptive and scalable solutions, motivating the exploration of learning-based approaches.

Limitations Identified
Scalability Optimization : Agentic architectures significantly increase computational complexity, limiting deployment in large-scale systems.
Learning-Driven Retrieval Policy : While adaptive and RL-based approaches show promise, stable and efficient policy learning remains underexplored.
Lack of Standardized evaluation Framework : There is no unified benchmarking methodology for measuring agentic autonomy, goal completion, and hallucination control across architectures.

TABLE III: Critical Identified Research Gaps

By synthesizing these observations, this study establishes a structured evaluation perspective on Agentic RAG methodologies. These insights motivate the need for adaptive, learning-driven coordination mechanisms. The subsequent section explores the integration of Reinforcement Learning within Agentic RAG frameworks, demonstrating how multi-level policy optimization can address existing limitations and enable scalable, efficient, and autonomous retrieval systems. To address these limitations, the following section introduces a reinforcement learning-based framework that enables adaptive and data-driven optimization of retrieval strategies.

J. Reinforcement Learning-Enhanced Agentic RAG Framework

To address the limitations identified in existing Agentic RAG methodologies, this paper proposes a Reinforcement Learning (RL)-enhanced Agentic RAG framework. The proposed approach introduces learning-driven optimization into the retrieval and reasoning pipeline, enabling dynamic adaptation, improved decision-making, and efficient multi-step task execution. Unlike existing approaches that rely on predefined rules or heuristics, the proposed framework introduces a learning-based mechanism for optimizing both planning and retrieval. The overall architecture of the proposed RL-enhanced Agentic RAG framework is illustrated in Fig. 3,

